

Intelligent Virtual Dental Implant Placement via 3D Segmentation Strategy

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Abstract

Virtual dental implant placement in cone-beam computed tomography (CBCT) is a prerequisite for digital implant surgery, carrying clinical significance. However, manual placement is a complex process that should meet clinical essential requirements of restoration orientation, bone adaptation, and anatomical safety. This complexity presents challenges in balancing multiple considerations comprehensively and automating the entire workflow efficiently. This study aims to achieve intelligent virtual dental implant placement through a 3-dimensional (3D) segmentation strategy. Focusing on the missing mandibular first molars, we developed a segmentation module based on nnU-Net to generate the virtual implant from the edentulous region of CBCT and employed an approximation module for mathematical optimization. The generated virtual implant was integrated with the original CBCT to meet clinical requirements. A total of 190 CBCT scans from 4 centers were collected for model development and testing. This tool segmented the virtual implant with a surface Dice coefficient (sDice) of 0.903 and 0.884 on internal and external testing sets. Compared to the ground truth, the average deviations of the implant platform, implant apex, and angle were 0.850 ± 0.554 mm, 1.442 ± 0.539 mm, and $4.927 \pm 3.804^\circ$ on the internal testing set and 0.822 ± 0.353 mm, 1.467 ± 0.560 mm, and $5.517 \pm 2.850^\circ$ on the external testing set, respectively. The 3D segmentation-based artificial intelligence tool demonstrated good performance in predicting both the dimension and position of the virtual implants, showing significant clinical application potential in implant planning.

Keywords: artificial intelligence, deep learning, dental prosthesis, decision making, computer-assisted surgery, cone-beam computed tomography

Introduction

With the advent of the digital era in oral implantology, implant surgical techniques have progressed from static templates to dynamic navigation and, more recently, to robot-assisted systems. These advancements have significantly enhanced surgical accuracy (Wang et al. 2024). Regardless of the surgical method, presurgical virtual implant placement remains a crucial step, serving as the “transmission protocol” that bridges imaging data with actual implant surgery. This step involves determining the appropriate implant dimensions and the optimal 3-dimensional (3D) position, adhering to clinical standards for restoration orientation, maximal utilization of residual bone, and avoidance of critical anatomical structures (Kalaivani et al. 2020; Chackartchi et al. 2022). However, balancing all these requirements is a complex, time-consuming, and error-prone task (Elgarba, Fontenele, Tarce, et al. 2024). Implantologists should first design the prosthesis, select a virtual implant with an appropriate dimension based on the available bone volume and anatomical constraints, and then place it in an optimal 3D position while integrating all these considerations (Alqutaibi et al. 2024). At present, virtual implant placement typically needs to be conducted or at least validated by experienced experts (Tatakis et al. 2019). Therefore, it remains a critical yet challenging clinical task, and its automation holds significant clinical importance (Appendix Fig. 1).

In recent years, artificial intelligence (AI) has rapidly advanced in implant dentistry (Revilla-León et al. 2023). However, there still exist several challenges in intelligent virtual dental implant placement. First, ideal implant placement should meet clinical standards for restoration orientation, bone adaptation, and anatomical safety while balancing these

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constraints optimally. However, these factors may conflict with others, particularly under complex clinical conditions (Ikbal et al. 2023; Dubey et al. 2024). Therefore, a significant challenge is to develop a model that can comprehensively consider multiple factors, ensuring sufficient robustness and generalization capability. Given the intricacy of implant planning, one possible approach involves training multiple networks to automate individual steps (including automatic implant crown design [Cho et al. 2024], important anatomical structure segmentation [Liu et al. 2024], and implant position prediction [Yang, Li, Li, Chen, et al. 2024; Yang, Li, Li, Wu, et al. 2024, etc.]) and then combine them to achieve full automation (Elgarba, Fontenele, Mangano, et al. 2024). Nevertheless, this approach may lead to inefficiencies and inaccuracies, as each step could introduce errors and require meticulous adjustment and validation, complicating the development and application of these methods. Thus, another challenge lies in simplifying the complex intermediate steps through a single network to enhance efficiency and reduce errors and computational burden.

A deep learning strategy leveraging a 3D image segmentation network offers a promising solution for intelligent virtual implant placement. These networks are capable of directly outputting the dimension and 3D position of target structures while robustly extracting features from cone-beam computed tomography (CBCT) images. On one hand, a 3D segmentation network can effectively analyze latent contextual information of the CBCT images through its unique network architectures. Therefore, through supervised learning, a 3D segmentation network is expected to learn the necessary decision factors for virtual implant placement and integrate them into the predictive outcomes. On the other hand, the dimension and position are the key information for virtual implant placement, while a 3D image segmentation network can classify and cluster voxels with the same semantics to determine the dimension and position of the objects (Li et al. 2024; Priya et al. 2024). Although there is no real implant in the input CBCT, the 3D image segmentation network may be able to directly predict the dimension and position of the virtual implant based on the context of the images, thus eliminating the need for complex intermediate steps.

Therefore, taking the mandibular first molar as an example, this study developed an intelligent virtual implant placement tool based on a 3D segmentation strategy. A segmentation model based on nnU-Net was constructed to generate the initial virtual implant mask from an edentulous region of interest (ROI). An approximation module was designed to mathematically optimize the segmentation mask, and a visualization module was designed to visualize the AI prediction results. Various metrics were used to evaluate segmentation performance and clinical feasibility of the proposed AI tool on multi-center testing sets. The hypothesis is that the proposed AI tool demonstrates strong performance in virtual implant placement (Fig. 1).

Materials and Methods

Data Set Development

This study conformed to the Declaration of Helsinki and was approved by the Medical Ethics Committee of Sun Yat-sen University Affiliated Hospital (No. KQEC-2020-29-06). Due to the retrospective nature of this study, informed consent was waived. A total of 190 CBCT scans were collected at the Department of Oral Implantology, Hospital of Stomatology, Sun Yat-sen University (center 1); Dongpu Clinic, Hospital of Stomatology, Sun Yat-sen University (center 2); Liangqinghua Dental Clinic (center 3); and Chunzhijia Dental Clinic (center 4). All CBCT devices were calibrated prior to scans. The acquisition parameters of the CBCT scans, including manufacturers, models, tube voltage, tube current, field of view (FOV), and voxel size, are described in Appendix Table 1A.

The inclusion criteria were as follows: 1) aged ≥ 18 y, 2) a single edentulous region in the mandibular first molar region (#36 or #46 tooth), and 3) adequate bone quantity to place an implant without bone augmentation. The exclusion criteria were as follows: 1) consecutive missing teeth in the mandibular posterior region; 2) CBCT images indicating mandibular cysts, tumors, or other pathological manifestations; 3) excessively tilted adjacent teeth resulting in significantly insufficient restorative space and making implant placement or prosthetic restoration infeasible; and 4) low-quality CBCT images with severe distortion, excessive noise, or any other issues that compromised the analysis required for dental implant planning.

Data Labeling and Preprocessing

This study established a standardized workflow for virtual implant placement (Appendix Materials and Methods). The manual virtual implant placement was restoration oriented and comprehensively considered both the bone volume and the position of the mandibular nerve canal. Virtual implant placements were performed by a specialist with over 10 y of clinical experience in dental implantology, using Mimics Research (version 21). The implant planning qualities were evaluated by 3 other specialists according to the established standards (Elgarba, Fontenele, Mangano, et al. 2024) (Appendix Table 2). To reduce the computational load, we determined the images with $96 \times 96 \times 96$ voxels from the original CBCT as edentulous ROIs. The standardized edentulous ROI determination and image preprocessing are described in the Appendix Materials and Methods.

Construction of the Intelligent Virtual Implant Placement Tool

A total of 174 CBCT scans from center 1 were used to develop the intelligent virtual implant placement tool. The scans were randomly assigned to a training set, validation set, and internal testing set (122, 35, and 17 CBCTs, respectively). The AI tool

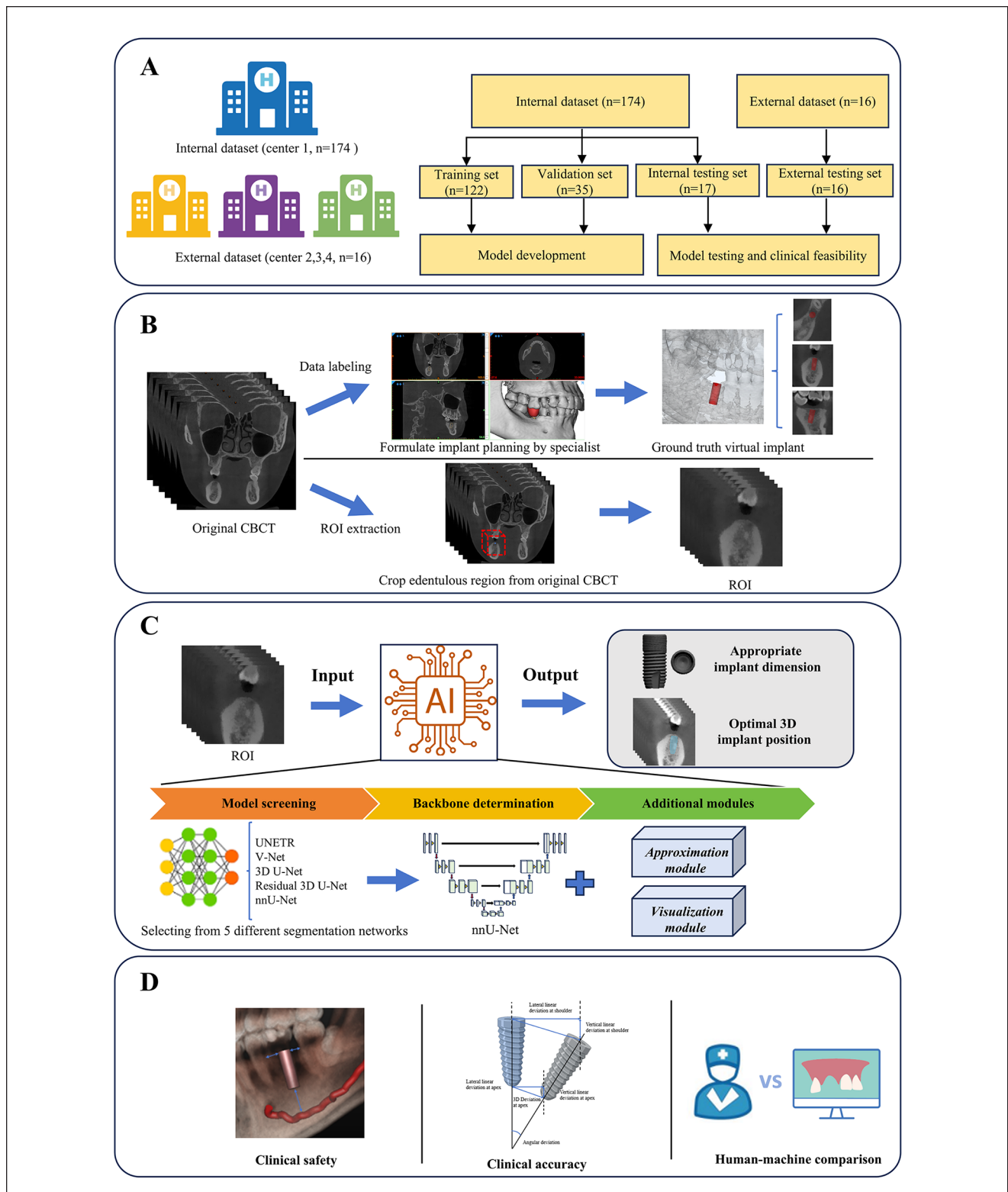


Figure 1. The flowchart of the development and validation of the intelligent virtual implant placement tool. **(A)** Cone-beam computed tomography (CBCT) scans were collected from 4 independent centers and divided into training, validation, internal testing, and external testing sets. **(B)** An experienced specialist manually formulated the implant planning to obtain the ground truth virtual implants. Edentulous regions with $96 \times 96 \times 96$ voxels were cropped from the original CBCT scans as regions of interest for model training. **(C)** Five 3D segmentation models were initially screened, and nnU-Net was selected as the backbone network with the best performance. Approximation and visualization modules were designed for postprocessing. These 3 modules constituted the intelligent implant planning tool. **(D)** The clinical feasibility and generalization ability of the AI tool were assessed on internal and external testing sets, including clinical safety, accuracy, and human-machine comparison.

consists of 3 modules, including the virtual implant segmentation module, the approximation module, and the visualization module. The development and training of the tool were performed on a local server equipped with NVIDIA GeForce RTX 3090 GPU.

The virtual implant segmentation module was a deep learning model to segment the primary virtual implant mask from the edentulous ROIs. Five distinct segmentation models (UNETR, 3D U-Net, V-Net, Residual 3D U-Net, and nnU-Net) were selected for training and validation. Tenfold data augmentation was applied by incorporating random biases and random flips. The patch size was $96 \times 96 \times 96$ voxels, and the batch size was set to 6. The Nesterov accelerated gradient optimizer and a combination of Dice loss and cross-entropy loss were used. The best-performing model was selected as the final backbone for the virtual implant segmentation module.

The approximation module was designed to fit the primary AI virtual implant mask into a regular cylinder, termed the final AI virtual implant. The approximation module first calculated the major axis and centroid of the primary virtual implant using the least squares method, then determined its length and radius. Based on these parameters, the module fitted the primary virtual implant into a regular cylinder to generate the final virtual implant.

The visualization module was designed to display the position of the virtual implant by integrating the AI virtual implant mask into the original CBCT. It modifies the Hounsfield unit (HU) values of the corresponding voxels in the original CBCT to 3,000, allowing the final AI virtual implant to appear as a white high-density shadow in the reconstructed CBCT.

Evaluation of the Intelligent Virtual Implant Placement Tool

The AI tool was evaluated on an internal testing set of 17 cases to assess segmentation performance, clinical safety, and accuracy. Additionally, an external testing set comprising 16 scans from centers 2 to 4 was used for further testing.

Segmentation performance. The segmentation performance of the model was evaluated using surface Dice coefficient (sDice), Dice coefficient, 95th percentile of the Hausdorff distance (HD95), and average surface distance (ASD) metrics. For the sDice, a tolerance of 1 mm was applied.

Clinical safety and accuracy. Clinical safety indexes were assessed by 2 well-trained implantologists through measuring the distances from the virtual implant margin to the buccal bone wall, lingual bone wall, adjacent teeth, and mandibular nerve canal (Appendix Fig. 3). The accuracy of AI virtual implant placement was evaluated by comparing the dimensions and positional deviations of the AI and ground truth (GT) virtual implants. Positional deviations include 3D, vertical linear, and lateral linear deviations at the implant shoulder and apex, as well as angular deviations (Appendix Fig. 4).

Human–Machine Comparison

To further evaluate the clinical applicability of the AI tool, 3 junior implantologists (with <5 y of clinical experience) and 3 senior implantologists (with >5 y of clinical experience) were invited to place virtual implants on the testing set. The positional deviations between the implants placed by the implantologists and the GT virtual implants were measured and compared with those of the AI virtual implants. The planning time required by each implantologist, from selecting the virtual implant to confirming its position, and the runtime of the AI model were recorded.

Statistical Analyses

Statistical analysis was conducted using SPSS (Version 25; IBM Corp.). The Kolmogorov–Smirnov test was used to assess the normality of the data. The intraclass correlation coefficient (ICC) was used to determine the consistency rate of the dimensions of the AI and GT virtual implants. Student's *t* tests and Wilcoxon signed-rank tests were employed to compare the statistical differences between positional deviation metrics of AI and GT virtual implants. The level of statistical significance was set at 0.05. ICC values greater than 0.75 indicate good agreement, and values between 0.5 and 0.75 indicate moderate agreement (Koo and Li 2016).

Results

Characteristics of the Multicenter Data

The study collected 190 CBCT scans (107 females and 83 males) from 4 centers, including 174 CBCT scans from an internal center and 16 CBCT scans from 3 external centers. A total of 190 virtual implants were placed as ground truth, with 90 in #36 tooth (47.3%) and 100 in #46 tooth (52.7%). The distribution of implant dimensions and cases under different clinical conditions is presented in Appendix Table 1. Three specialists evaluated the GT implant planning qualities with scores of 4.65 ± 0.49 , 4.45 ± 0.51 , and 4.50 ± 0.51 , with an average score of 4.53 ± 0.42 .

Construction of the Intelligent Virtual Implant Placement Tool

The nnU-Net architecture performed the best among 3D segmentation networks, achieving a sDice of 0.906 and Dice of 0.727 on the validation set (Appendix Table 3), and was therefore selected as the final backbone for the virtual implant segmentation module. After 500 epochs, the network converged in both training and validation sets (Appendix Fig. 5).

The final workflow of the intelligent implant planning tool is illustrated in Figure 2A. When inputting the edentulous ROI, the tool automatically outputs the virtual implant with the appropriate implant dimension and optimal 3D position.

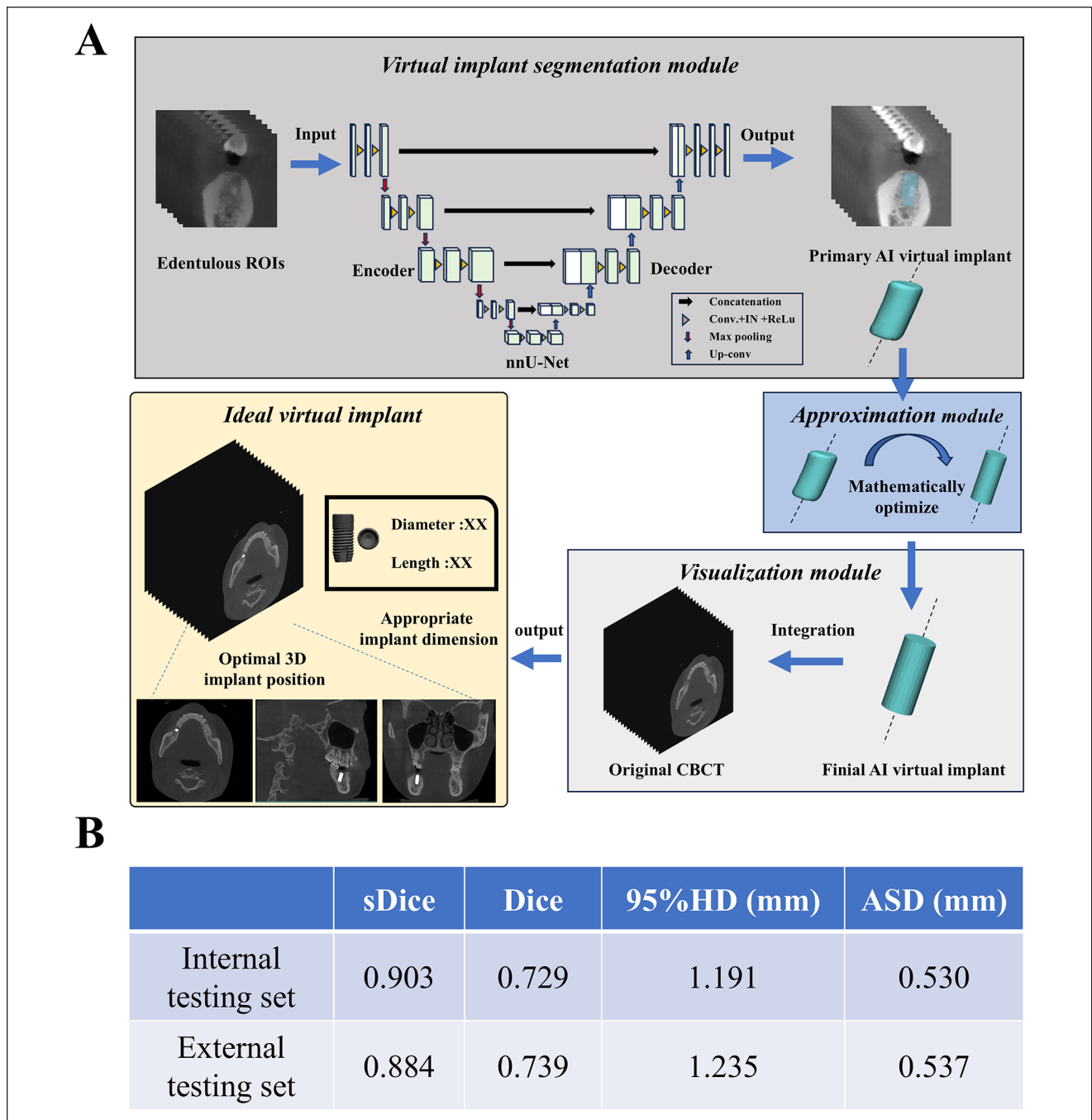


Figure 2. The schematic of the workflow of the intelligent virtual implant placement tool and its segmentation performance. **(A)** The model comprises 3 modules, including the virtual implant segmentation module, approximation module, and visualization module. First, the edentulous region was input into the virtual implant segmentation module based on nnU-Net to segment the primary virtual implant mask. The approximation module then mathematically optimized the primary artificial intelligence (AI) virtual implant as a circular cylinder to obtain the final AI virtual implant mask. Finally, the visualization module integrated the final AI virtual implant with the original cone-beam computed tomography to visualize the virtual implant. **(B)** The performance of the AI tool for virtual implant segmentation on the internal and external testing sets.

Evaluation of the Intelligent Virtual Implant Placement

Segmentation performance. The sDice, Dice, HD95, and ASD for the virtual implant segmentation were 0.903, 0.729, 1.191 mm, and 0.530 mm on the internal testing set and 0.884, 0.739, 1.235 mm, and 0.537 mm, respectively, on the external

testing set (Fig. 2B). Visualization segmentation results of 8 randomly selected cases showed the predominance of green in the heatmap, indicating the AI virtual implants closely resembled the GT (Fig. 3A).

Clinical safety and accuracy. There was no significant difference in the average distances between the AI and the GT

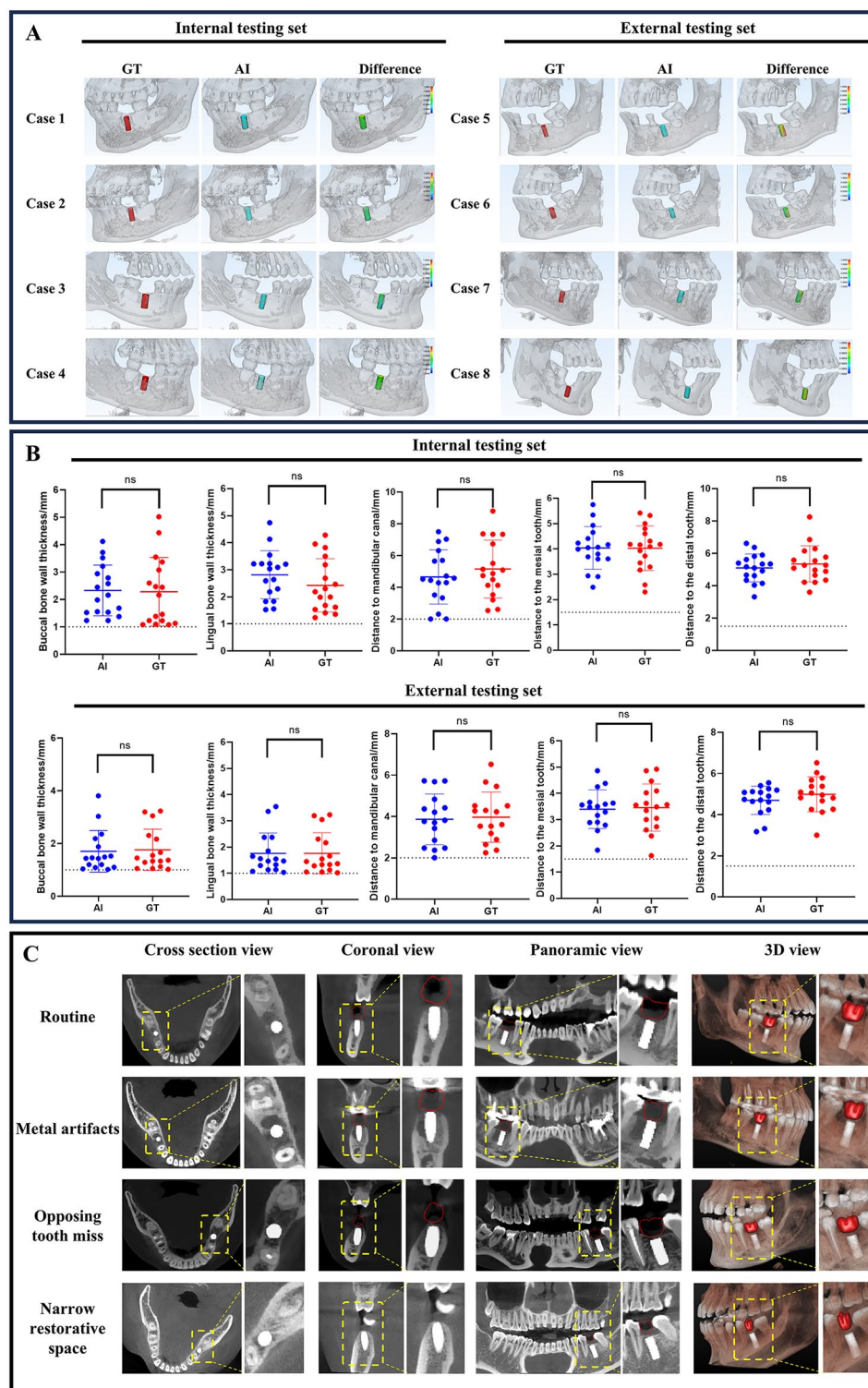


Figure 3. The performance of the intelligent virtual implant placement tool. **(A)** The visualization of the artificial intelligence (AI) and ground truth (GT) virtual implants. The error between the AI virtual implants (blue) and the GT virtual implants (red) is displayed through a heatmap. The predominance of green in the heatmap indicates that the AI virtual implants closely match GT virtual implants, both on the internal and external testing sets. **(B)** The scatterplots of the clinical safety evaluation of AI and GT virtual implants. The AI and GT virtual implants all met the safety requirements on both the internal and external testing sets. **(C)** Demonstration of the intelligent implant planning results under different clinical conditions. Under both routine and challenging clinical conditions, all the AI virtual implants were aligned with the axis of the virtual crown and maintained appropriate distances from the buccal and lingual bone walls, adjacent tooth roots, and the mandibular nerve canal. ns, not significant.

Table. Clinical Acceptance of the AI Virtual Implant on Internal and External Testing Sets.

	Internal Testing Set	External Testing Set	P Value
Consistency of the dimension between the AI and GT virtual implants			
ICC			
Implant length	0.867	0.799	
Implant diameter	0.590	0.784	
Linear and angular deviations between AI and GT virtual implants			
Position deviation indexes			
3D deviation at shoulder (mm)	0.850 ± 0.554	0.822 ± 0.353	0.533 ^a
Vertical linear deviation at shoulder (mm)	0.389 ± 0.364	0.372 ± 0.339	0.839 ^a
Lateral linear deviation at shoulder (mm)	0.715 ± 0.488	0.676 ± 0.309	0.728 ^a
3D deviation at apex (mm)	1.442 ± 0.539	1.467 ± 0.560	0.894 ^b
Vertical linear deviation at apex (mm)	0.708 ± 0.517	0.812 ± 0.607	0.228 ^b
Lateral linear deviation at apex (mm)	1.116 ± 0.613	1.077 ± 0.550	0.728 ^a
Angular deviation (degree)	4.927 ± 3.804	5.517 ± 2.850	0.873 ^b

AI, artificial intelligence; GT, ground truth; ICC, intraclass correlation coefficient; 3D, 3-dimensional.

^aWilcoxon signed-rank tests.

^bStudent's *t* tests.

virtual implants from critical anatomical structures. All virtual implants met the clinical safety standard (Fig. 3B).

The length consistency between the AI and GT virtual implants exhibited good agreement on both internal and external testing sets. The diameter consistency demonstrated moderate agreement on the internal testing set but achieved good agreement on the external testing set (Table). In terms of placement positions, the deviations in implant platform, implant apex, and angle between the AI and GT virtual implants were 0.850 ± 0.554 mm, 1.442 ± 0.539 mm, and 4.927 ± 3.804 degrees, respectively, on the internal testing set and 0.822 ± 0.353 mm, 1.467 ± 0.560 mm, and 5.517 ± 2.850 degrees, respectively, on external testing set. There was no significant difference in implant position deviation between the internal and the external testing sets ($P > 0.05$) (Table).

The performance of intelligent implant planning under both routine and challenging clinical conditions is shown in Figure 3C. The AI virtual implants were aligned with the axis of the crowns predesigned by the implantologist and maintained appropriate distances from the adjacent anatomical structures.

Human–Machine Comparison

Compared to junior implantologists, the AI virtual implants demonstrated significantly lower positional deviations across almost all metrics, and only a few differences were observed compared to senior implantologists. The AI model required significantly less time than implantologists for each patient (Fig. 4). This demonstrated that the AI tool has high robustness.

Discussion

Construction of the Intelligent Virtual Implant Placement Tool

In this study, we constructed a segmentation model to predict the dimensions and positions of virtual implants based on contextual information from CBCT scans, achieving automatic intelligent

virtual implant placement. Previous studies tended to obtain relevant information about implants by defining the structures, which may limit models' performance under complex clinical conditions (Chen et al. 2024). In fact, the GT virtual implant can represent an expert-level compromise among various complex decision factors (Chackartchi et al. 2022). These decision factors are contained in the latent contextual information of the CBCT images. Therefore, instead of relying on predefined structural components or explicitly defined constraints, this study adopted an approach where virtual implant placement was implicitly learned from implant planning data sets. This methodology enhances the robustness of our AI tool, enabling it to remain effective under challenging clinical conditions (Fig. 3C).

The key to achieving intelligent virtual implant placement is constructing a model that can comprehensively analyze CBCT information and accurately predict the mask. The outstanding performance of nnU-Net may be attributed to its special architecture: the encoder–decoder structure and skip connections allow for the fusion of semantic features, the adaptive parameter framework optimizes model configuration, and layer-by-layer cascading of feature maps enhances prediction accuracy (Isensee et al. 2021). Unlike fixed-architecture models, such as V-Net, nnU-Net can dynamically adjust hyperparameters, significantly improving feature extraction and prediction capabilities. Compared to Transformer-based architectures, such as UNETR, nnU-Net demonstrates superior performance in scenarios with limited training data and computing resources (Azad et al. 2024; Zhang et al. 2024). Additionally, the approximation module leverages the prior knowledge that implants are approximately cylindrical, preventing the generation of irregularly shaped implants, thereby making our tool more suitable for clinical applications.

The visualization module can integrate the AI virtual implant with the original CBCT and save the reconstructed CBCT as DICOM files. Due to the universality of the DICOM format, the reconstructed CBCT can be viewed in almost all surgical template, navigation, and implant-robot scheme design software (D'Addazio et al. 2022; Zhao et al. 2024).

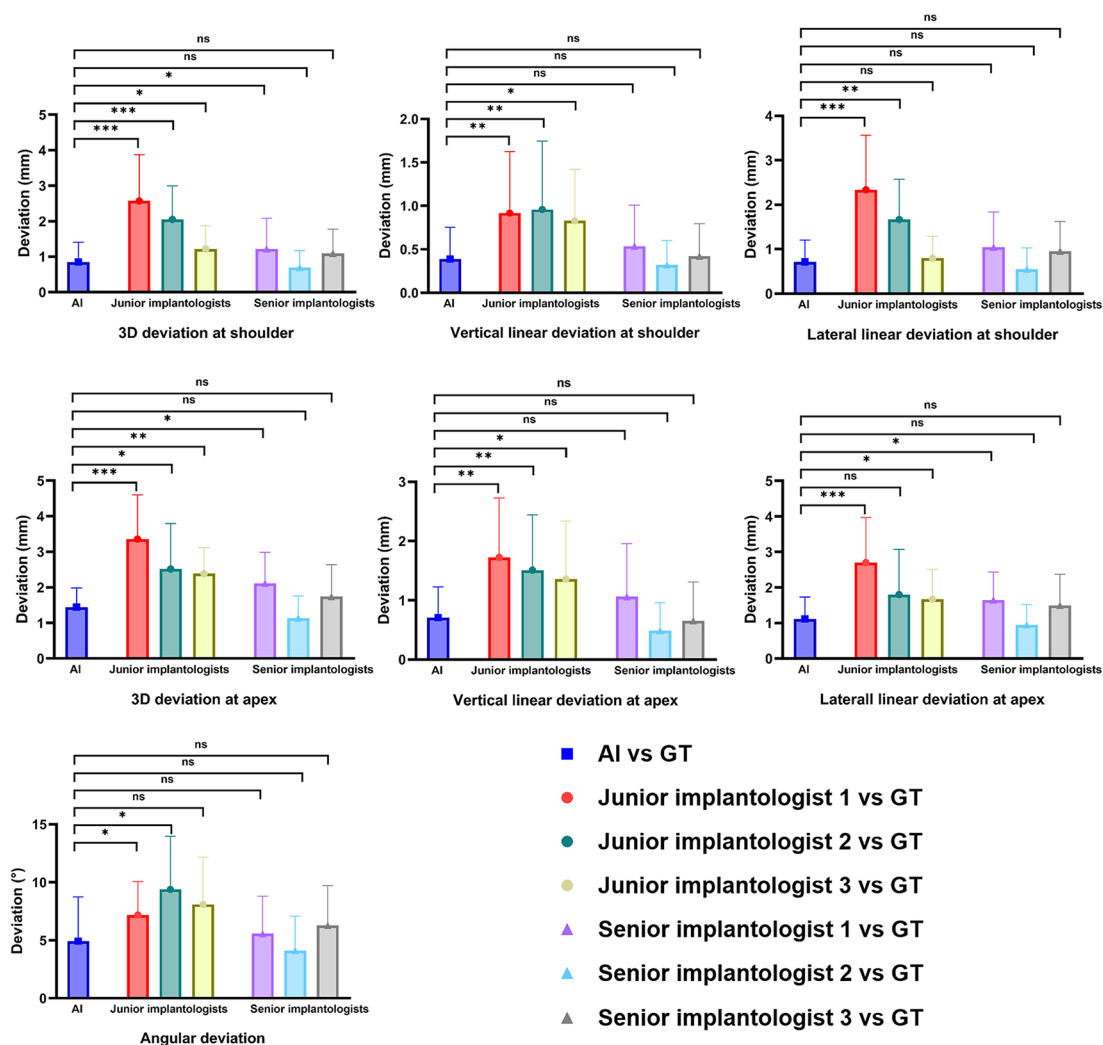


Figure 4. The comparison of position deviation and efficacy between artificial intelligence and implantologists in implant planning.
 * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

Therefore, implantologists can quickly complete digital implant schemes based on the reconstructed CBCT.

Evaluation of the Intelligent Virtual Implant Placement Tool

Evaluating the implant planning tool at both the segmentation capability and clinical levels is crucial. The sDice can reflect the segmentation performance within a given tolerance range by treating surfaces within this tolerance as overlapping parts and then calculating the Dice coefficient (Walter et al. 2024). Considering that implant planning is subjective and allows for slight deviation, the sDice was used as the primary evaluation metric for segmentation ability, which helped reduce bias and make the evaluation criteria more aligned with clinical requirements. Since the clinically permissible implant tolerance is about 1 mm, the sDice tolerance was set to 1 mm (Yang, Li, Li, Wu, et al. 2024). The sDice on the internal and external testing sets were 0.903 and 0.884, respectively, indicating that the model has satisfactory performance.

As a clinically oriented intelligent tool, safety is the top priority. Our tool achieved satisfactory clinical safety results, showing no invasion of important anatomical structures (Fig. 3B). Moreover, clinical placement accuracy should be evaluated. Although the deviations of our AI tool are equivalent to implant placement with a fully surgical template-guided approach and navigation approach, it is important to note that the deviations of the AI tool are still at the presurgical design stage (Wei et al. 2021; Aghaloo et al. 2023). If the AI is integrated with those digital surgical execution tools, potential deviation accumulation may occur due to factors such as surgical technique, bone density, and patient-specific anatomical variations. For example, a study has reported that different bone densities can affect the deviation in AI-assisted design of implant guides (Chen et al. 2024). Therefore, it is necessary to further analyze the integration of AI dental planning tools with digital surgical execution tools to ensure the accuracy of the final implant position and AI design position.

Strong generalization is crucial for an AI model to be applicable in real-world scenarios (Cui et al. 2022; Ni et al. 2024). Most studies on AI implant planning used a single-center data set, without verification of model generalization ability in multi-center data sets (Moufti et al. 2023; Cho et al. 2024; Feng et al. 2024). In this study, we created an external test set from a multi-center cohort for evaluating the generalization ability. Our model demonstrated comparable performance on the external data set as on the internal data set, proving its generalization capability (Table). Additionally, due to limited clinical experience, junior implantologists may introduce more errors in implant planning. The AI deviations were nearly identical to those of senior implantologists, suggesting that its performance is comparable to that of experienced professionals (Fig. 4). Therefore, this AI tool has the potential to serve as a high-quality reference, assisting clinicians in achieving more accurate and efficient implant planning.

Our study has some limitations. First, our research focused only on single tooth loss and did not address more complex

clinical situations such as edentulous arches. Second, the sample size remains relatively small and is from a limited expert source. Future work will extend the model to complex implant planning within this research framework and enhance generalizability by incorporating larger-scale, multicenter, and multi-expert source data sets. Additionally, prospective clinical trials are required to further validate the tool's clinical utility.

Conclusions

This study proposed a novel intelligent strategy for virtual implant placement and successfully applied it to the missing mandibular first molar. The tool demonstrated strong performance in predicting both the dimension and 3D position of the virtual implant. This approach has significant clinical potential and can serve as a template for addressing complex intelligent implant planning.

Author Contribution

G. Cai, contributed to conception, design, data acquisition, analysis, and interpretation, drafted and critically revised the manuscript; B. Wen, contributed to conception, design, data analysis and interpretation, drafted and critically revised the manuscript; Z. Gong, contributed to conception, design, data acquisition and analysis, drafted and critically revised the manuscript; Y. Lin, contributed to conception, data acquisition, drafted the manuscript; H. Liu, contributed to conception, data interpretation, critically revised the manuscript; P. Zeng, contributed to design, data analysis, critically revised the manuscript; M. Shi, contributed to conception, design, data acquisition and interpretation, critically revised the manuscript; R. Wang, Z. Chen, contributed to conception, design, data analysis and interpretation, critically revised the manuscript. All authors gave final approval and agree to be accountable for all aspects of the work.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Code and Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. The code is released at <https://github.com/wenbc21/ImplantGenerator>.

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